

Research Statement

This statement outlines my research contributions, encompassing both published work and ongoing projects. It also includes an overview of my future research agenda

Research Interests and Positioning

My research occupies the critical intersection of Strategic Management, Innovation, Economics, and Computational Social Science. Theoretically, my work is deeply rooted in the micro-foundations of strategy, specifically examining how individual behaviors, cognitive frameworks, and interpersonal relationships scale up to shape firm-level strategic choices and outcomes as well as, platform governance, and broader market outcomes. A central, unifying theme of my research agenda is understanding how emerging technologies and digital architectures reshape interpersonal and inter-firm dynamics. As digital platforms and algorithmic governance replace traditional organizational boundaries, they fundamentally alter classic strategic trade-offs between competition and collaboration (co-opetition).

To study these complex, multi-level behaviors, ranging from relational rivalry and structural sabotage to platform governance for ecosystem tax compliance, I combine rigorous quasi-experimental designs with advanced computational methods, including web-scraping, machine learning, and large language models (LLMs). I focus on research projects that have the potential to be published in high-quality journals and have published papers in Management Science and Organization Science. For my research projects, I was able to attract significant external funding (CHF ~450,000) together with my co-authors.

My research is aligned with DORA principles and a commitment to Open Science. As a member of the Management Science Reproducibility Team, I participated in the journal's efforts to verify and improve the replicability of peer-reviewed research. Furthermore, my current role on the Copenhagen Business School (CBS) Steering Committee on Research Data Management involves shaping institutional frameworks for data transparency, security, and reproducible workflows.

Research streams

All my research aims at using large-scale data and rigorous empirical designs to study causal effects. However, broadly my research can be divided into at least two different streams:

My first research stream investigates how individuals navigate high-stakes, competitive environments, isolating the behavioral mechanisms and interpersonal frictions that drive or destroy performance. By examining the micro-foundations of competitive interactions, this work uncovers how psychological and relational constructs impact broader organizational systems and performance outcomes. In a paper published at Management Science, my co-authors and I challenge the prevailing assumption in management research that rivalry is purely performance-enhancing. Utilizing a quasi-experimental design tracking 4.6 million competitive encounters in an online coding platform, we uncover the negative performance consequences of deep relational rivalry. We show

that for low-skilled individuals and those at risk of status loss exposure to high levels of rivalry can severely backfire, degrading cognitive performance and altering strategic risk-taking. In another paper, my co-authors and I explore the structural tension between collaboration and competition within crowdsourcing and peer-based corporate communities. We derive formal predictions from a game-theoretic model and test them against a massive empirical dataset of 38 million peer-evaluations. The paper sheds light on how strategic actors balance collective success against individual positioning, capturing otherwise covert behaviors such as systemic sabotage and strategic self-promotion and how these individual behaviors collectively affect platform level outcomes.

My second research stream focuses on how platform architectures and design choices govern economic behavior and distribute value within digital ecosystems. In a study published in *Management Science*, my co-authors and I analyze how platform design changes the dynamics between types of platform complementors, in particular professional producers and citizen-generated content. For that I trained a custom language model on over 2 million new articles to track content shifts and strategic spillovers. The findings demonstrate how digital architectures can help mitigate the emergence of local informational deserts, showcasing how technological platforms can be strategically steered to alter market-wide outcomes. In another project, which has been supported by a DKK 2.8 million grant from the Rockwool Foundation and is currently prepared for submission to *SMJ*, we investigate the strategic boundary between private platform governance and public regulation. Merging global platform data (Airbnb) with highly granular administrative registries, we exploit a natural experiment to study how different types of collaboration between platform owners and regulators in the form of different information-sharing agreements alter on-platform complementor behavior and off-platform regulatory compliance. In a third project in this stream, we analyze the role of non-monetary, symbolic rewards for platform governance in firm-hosted online communities. Using GPT-4 to run automated text and behavioral classifications in a quasi-experimental layout across platforms, we study how the introduction of performance-contingent symbolic awards by the platform owner clashes with the established incentive norms of participation and erode contributions to online common goods. In a related project, currently being prepared for *Research Policy*, my coauthors and I study the diffusion consequences of motivational profiles. Specifically, we explore how different types of individual motivation affect the diffusion of user innovations in social movements. We deploy machine learning to analyze 69 million social media posts and combine it with granular survey data to map how users structurally differ in how they diffuse their knowledge across ecosystem boundaries.

Future agenda

My future research agenda sits at the intersection of behavioral strategy and digital innovation, focusing on how structural shifts in technology alter multi-level competitive dynamics, firm innovation, and individual well-being.

First, in a larger project around the role of AI for firm and individual competition, I want to explore multiplex competition, where e.g. individuals compete internally for promotions while their firms

compete for market share, and how these dynamics change with the introduction of AI. Generative AI introduces a disruptive third layer, forcing knowledge-workers to compete against technological substitutes. By deploying NLP to scrape firm websites for real-time market positioning and AI adoption, and linking these to employer-employee registries and medical histories, this project causally estimates how technological substitution interacts with traditional rivalries to generate workplace techno-stress or fuel strategic adaptation of firms and individuals. This directly addresses how to design resilient competitive environments at the intersection of firms, individuals, and technologies.

In a second stream I want to investigate the strategic dilemma firms face when evaluating the quality of external technologies comparing proprietary and open-source technologies. While open-source ecosystems reduce validation frictions by making quality signals publicly observable, they shift the burden of interpretation and applicability onto the adopting firm. Empirically, I plan to construct a longitudinal panel tracking web-stack tech choices across hundreds of thousands of domains and matching them to firm-level register information. By leveraging precisely dated software vulnerabilities as exogenous negative shocks to technology quality, this stream investigates the internal organizational capabilities required to successfully manage technology-driven innovation.

Methodologically, the work leverages advanced natural language processing, large language models, and large-scale digital trace data linked to administrative registers. For these projects I intend to secure funding from sources such as e.g. the Swiss National Science Foundation or the ERC.